

Unlocking the Potential of Single-Cell Omics

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The cellular, molecular, and functional diversity of complex tissues has motivated sustained research efforts toward developing methodologies for quantifying this diversity. Much of the investments and efforts have focused on analyzing nucleic acids using second generation sequencing methods, and this has resulted in large scale cellular atlases of healthy and diseased tissues. Despite these advances in describing cellular compositions, identifying molecular mechanisms based on nucleic acid analysis alone remains challenging. Indeed, individual human cells regulate their fates post-transcriptionally,^{1,2} and thus studying such processes can be greatly enabled by single-cell proteomics.³

The aim for mechanistic understanding motivates methodological expansion of single-cell analysis toward proteins, metabolites, and lipids. The *Journal of Proteome Research* has long facilitated the dissemination of such methods and highlighted opportunities at key points of technology development.⁴ This **Special Issue on Single-Cell Omics** serves this mission by collating diverse articles. They include methods for increasing the sensitivity of protein quantification with isobaric mass tags, characterizing the impact of cell storage conditions, quantifying lipids in single cells, and perspectives on single-cell metabolomics and practical scalability of single-cell proteomics by mass spectrometry. The increasing scope and power of single-cell omics methods open the potential for previously inaccessible biological investigations, such as quantitative interpretations of protein covariation across single cells.⁵ Making the most of this potential, requires good experimental designs addressing well-posed biomedical questions.

To maximize the utility of single-cell methods and analysis, the data and metadata should be shared in standardized formats with robust quality controls according to community guidelines.⁶ Adhering to such best practices and updating them as needed contributes to reliable scientific results and healthy growth of the field. The *Journal of Proteome Research* seeks to encourage such healthy growth by continuing to disseminate new methods and research in the rapidly advancing field of single-cell omics.

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Notes

Views expressed in this editorial are those of the author and not necessarily the views of the ACS.

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